Started on	Sunday, 24 July 2022, 4:50 PM
State	Finished
Completed on	Sunday, 24 July 2022, 4:51 PM
Time taken	10 secs
Marks	0.00/51.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Approximately what percentage of epileptic patients will have normal EEG between attacks?

Select one:

- 70-90%
- 1-2%
- 10-15%
- 30-50%
- 5-10%

Explanation:

When an EEG is performed in the outpatient setting, it detects epileptiform abnormalities in only 29-55% of patients in any type of focal epilepsy.

Ref: Boutros NN, Galderisi S, Pogarell O, Riggio S. Standard electroencephalography in clinical psychiatry: A practical handbook. 2011. pp. 86-87.

The correct answer is: 30-50%

Question **2**

Not answered

Marked out of 1.00

Flag question

Flattened trace of EEG is a feature seen in

Select one:

- Petit mal epilepsy
- Creutzfeldt-Jakob disease
- Delirium
- Partial seizures
- Huntington disease

A significant reduction in voltage with a predominant absence of alpha rhythms, sometimes leading to flat traces, is seen in Huntington disease.

The correct answer is: Huntington disease

Question **3**

Not answered

Marked out of 1.00

Flag question

REM sleep includes which of the following physiological features?

Select one:

- Abolition of tendon reflexes
- Increased parasympathetic activity
- Reduced heart rate
- Increased recall of dreams if awoken
- Reduced cerebral blood flow

Explanation:

Because most people awakened during a REM episode recall that they were dreaming, physicians equate REM sleep with dreaming. Dreams that occur during REM sleep possess emotional or intellectual complexity, at least on a superficial level, and rich visual imagery that people readily recall. During REM sleep, in contrast to the accompanying flaccid muscle paralysis, autonomic nervous system activity increases and produces an increased pulse rate, elevated blood pressure, raised intracranial pressure, increased cerebral blood flow, greater muscle metabolism, and, in boys as well as men, erections.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 379.

The correct answer is: Increased recall of dreams if awoken

Question **4**

Not answered

Marked out of 1.00

Flag question

The EEG wave with a frequency of 4 to 8 Hz is

Select one:

- Theta
- Delta
- Alpha
- Gamma
- Beta

Explanation:

Alpha waves: frequency range 8-13 Hz; highly rhythmic and constitute the dominant brain wave frequency of the normal eyes-closed awake EEG

Beta waves: frequencies that are faster than the upper 13 Hz limit of the alpha rhythm; not uncommon in average adult waking EEG studies, particularly over the frontal-central regions

Delta waves: frequency of 3.5 Hz or less; not present in the normal waking EEG, but a prominent feature of deeper stages of sleep; presence of significant generalised or focal delta waves in the awake EEG is strongly indicative of a pathophysiological process

Theta waves: frequency 4-7.5 Hz; small amount of sporadic, arrhythmic, and isolated theta activity can be seen in many normal waking EEG studies, particularly in frontal-temporal regions; a prominent feature of the drowsy and sleep tracing; excessive theta in awake EEG, generalised, or focal, suggests

the operation of a pathologic process

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 998-999.

The correct answer is: Theta

Question **5**

Not answered

Marked out of 1.00

Flag question

A patient is administered 1g of dexamethasone at 11PM, and his plasma samples are collected to measure hormone levels on the following morning. Which of the following is correct concerning the physiological changes overnight?

Select one:

- ACTH levels are suppressed
- CRH levels are increased
- Glucocorticoid receptors are blocked
- Prolactin levels are increased
- Cortisol levels are increased

The dexamethasone suppression test (DST) is used to demonstrate a failure of feedback suppression of ACTH/CRH and continuous production of endogenous cortisol despite administration of exogenous steroid (dexamethasone). Normally administering dexamethasone must reduce cortisol in plasma. This is because of intact HPA function leading to reduced ACTH and CRH. In depression and other psychiatric hypercortisolemic states (also in organic hypercortisolemic states such as Cushing's), this does not occur.

The correct answer is: ACTH levels are suppressed

Question **6**

Not answered

Marked out of 1.00

Flag question

The resting membrane potential of a neuron is around

Select one:

- 170 mV
- 70 mV
- 30 mV
- -30 mV
- -70 mV

Explanation:

A neural membrane at resting potential is about -70mV (meaning that the intracellular side of the membrane is -70mV with respect to the extracellular environment). This resting potential is caused by differences in the relative concentration and permeability of various inorganic ions, particularly sodium, potassium, and chloride, as well as by the active contributions of a sodium-potassium pump.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. p. 86.

The correct answer is: -70 mV

Question **7**

Not answered

Marked out of 1.00

Flag question

Which of the following is associated with low levels of cortisol?

Select one:

- PTSD
- Depression
- Anorexia
- Psychosis
- Mania

Explanation:

Although increased cortisol levels are typically associated with chronic stress, PTSD is often linked with lower cortisol levels. Further, lower cortisol levels shortly after trauma predict subsequent PTSD severity. This paradoxical finding has been interpreted in term of cortisol binding to the glucocorticoid receptors in a negative feedback loop that promotes homeostasis of the stress response. This proposal posits that lower cortisol in PTSD may result in elevated ongoing activity of the hypothalamic-pituitary-adrenal (HPA) axis, resulting in exaggerated catecholamine response and subsequent over-consolidation of trauma memories.

Ref: Bryant RA. Post-traumatic stress disorder: a state-of-the-art review of evidence and challenges. World Psychiatry. 2019 Oct;18(3):259-69.

The correct answer is: PTSD

Question **8**

Not answered

Marked out of 1.00

Flag question

EEG changes in Angelman Syndrome are notable by the age of

Select one:

- Soon after birth
- Two years
- Puberty
- 7 years
- 21 years

Explanation:

Angelman Syndrome (AS) is characterised by severe developmental delays including marked speech impairment, movement abnormalities, (ataxia, tremor), and unique behaviours such as frequent laughter. Intractable epileptic seizures since early childhood with characteristic EEG abnormalities are present in 80-90% of patients with AS. The onset of epilepsy is most prevalent between 1 and 3 years of age.

Ref: Samanta D. Epilepsy in Angelman syndrome: A scoping review. Brain and Development. 2020 Sep 4;1-13.

The correct answer is: Two years

Question **9**

Not answered

Marked out of 1.00

Flag question

The frequency of ß waves seen in normal EEG is

Select one:

- 8-13 Hz
- More than 40Hz
- 4-8Hz
- Less than 7 Hz
- More than 13Hz

Explanation:

EEG Rhythm	Frequency range	Amplitude	Main scalp area	Subject condition
Delta	0.1-3.5 Hz	50-350 uV	Variable	Drowsiness and sleep Hyperventilation Infancy and childhood
Theta	4-7.5 Hz	10-150 uV	Variable	Drowsiness and sleep Hyperventilation Infancy and childhood Small amount in awake adults
Alpha	8-13 Hz	20-100 uV	Posterior (occipital-parietal)	Relaxed wakefulness with eyes closed
Beta	Above 13 Hz	10-30 uV	Frontal or diffuse	Increase during cognitive efforts as well as during drowsiness and sleep
Mu	7-12 Hz	10-50 uV	Central	Relaxed wakefulness with both eyes open and closed Blocked by movements of the contralateral body parts
Lambda	Sharp transients of 200-300 msec duration	Below 50 uV	Occipital	Visual exploration

Adapted from Standard electroencephalography in clinical psychiatry: A practical handbook. 2011. p. 34.

SPMM COURSE

Ref: Boutros NN, Galderisi S, Pogarell O, Riggio S. Standard electroencephalography in clinical psychiatry: A practical handbook. 2011. p. 34.

The correct answer is: More than 13Hz

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following physiological changes are noted after ECT in clinically depressed individuals?

Select one:

- Reduced noradrenaline turnover
- Beta waves predominance in EEG
- Reduced ß receptors
- Increased a2 receptors
- Increase in 5HT2 receptors

Explanation:

Virtually every neurotransmitter system is affected by ECT, but a series of ECT sessions results in the

downregulation of post-synaptic beta-adrenergic receptors, the same receptor change observed with virtually all antidepressant treatments. The effects of ECT on serotonergic neurons remain controversial.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 731.

The correct answer is: Reduced ß receptors

Question **11**

Not answered

Marked out of 1.00

Flag question

Regular 3 Hz complexes are seen in EEG of patients with which of the following conditions?

Select one:

- ADHD
- CJD
- Antisocial personality disorder
- Angleman syndrome
- Absence seizures

Explanation:

In patients with absence seizures (petit mal), the EEG produces a characteristic pattern of three-per-second spike-and-wave activity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 250.

The correct answer is: Absence seizures

Question **12**

Not answered

Marked out of 1.00

Flag question

Which one of the following statements is true about normal EEG?

Select one:

- The normal dominant alpha rhythm is usually achieved by 8-10 years old.
- Closing the eyes usually abolishes the dominant alpha rhythm.
- Ongoing delta activity when awake is almost always abnormal in the awake adult.
- Beta activity is almost always abnormal in the adult.
- The dominant alpha rhythm is seen in the fronto-temporal region.

Explanation:

By mid-adolescence, the EEG essentially has the appearance of an adult tracing.

Alpha waves (frequency range 8-13 Hz) are highly rhythmic and constitute the dominant brain wave frequency of the normal eyes-closed awake EEG.

Beta waves (frequencies that are faster than the upper 13 Hz limit of the alpha rhythm) are not uncommon in average adult waking EEG studies.

Delta waves (frequency of 3.5 Hz or less) are not present in the normal waking EEG, but a prominent feature of deeper stages of sleep. Presence of significant generalised or focal delta waves in the awake EEG is strongly indicative of a pathophysiological process.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp . 998-

999.

The correct answer is: Ongoing delta activity when awake is almost always abnormal in the awake adult.

Question **13**

Not answered

Marked out of 1.00

Flag question

Which of the following sleep EEG changes is seen when taking benzodiazepines for insomnia?

Select one:

- Increased slow wave sleep
- Reduced stage 1 sleep
- Reduced stage 2 sleep
- Increased REM sleep
- Reduced REM latency

Multiple sleep-related changes including a decrease in sleep latency, increased sleep time, reduced stage 1 sleep, increased stage 2 sleep, reduced REM and Slow Wave Sleep, and REM rebound on cessation are seen with benzodiazepines.

The correct answer is: Reduced stage 1 sleep

Question **14**

Not answered

Marked out of 1.00

Flag question

An EEG showing high amplitude repetitive bilaterally synchronous symmetrical polyphasic sharp wave and slow wave complexes, which occur every 4-15 seconds are characteristic of

Select one:

- Neurosyphilis
- Variant Creutzfeldt-Jakob disease
- Herpes simplex encephalitis
- Subacute sclerosing panencephalitis
- Absence seizures

Explanation:

Subacute sclerosing panencephalitis (SSPE) is a rare, predominantly childhood-onset neurologic illness that probably results from a latent or mutant measles (rubella) virus infection of neurons. Alternatively, the body's aberrant response to this infection causes the illness. Clinicians can confirm a diagnosis of SSPE by finding an elevated measles titre in the CSF. An EEG is also diagnostically helpful: it shows periodic sharp-wave complexes or a burst-suppression pattern. Generalised periodic discharges appear as regular, sharp waves or spikes. They can have a clinical correlate in myoclonic jerks. The EEG pattern of Creutzfeldt-Jakob disease (CJD) also shows periodic discharges. Unlike conventional CJD which typically appears at age 50-64 years, variant CJD (vCJD) typically affects teens and young adults, and the EEG lacks periodic complexes.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 131-132, 154, 210.

The correct answer is: Subacute sclerosing panencephalitis

Question **15**

Not answered

Marked out of 1.00

Flag question

The delta waves have a frequency of

Select one:

- 13-40 Hz
- 8-13 Hz
- 3.5-7.5 Hz
- 7-11 Hz
- Less than 4 Hz

Explanation:

Alpha waves: frequency range 8-13 Hz; highly rhythmic and constitute the dominant brain wave frequency of the normal eyes-closed awake EEG

Beta waves: frequencies that are faster than the upper 13 Hz limit of the alpha rhythm; not uncommon in average adult waking EEG studies, particularly over the frontal-central regions

Delta waves: frequency of 3.5 Hz or less; not present in the normal waking EEG, but a prominent feature of deeper stages of sleep; presence of significant generalised or focal delta waves in the awake EEG is strongly indicative of a pathophysiological process

Theta waves: frequency 4-7.5 Hz; small amount of sporadic, arrhythmic, and isolated theta activity can be seen in many normal waking EEG studies, particularly in frontal-temporal regions; a prominent feature of the drowsy and sleep tracing; excessive theta in awake EEG, generalised, or focal, suggests the operation of a pathologic process

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 998-999.

The correct answer is: Less than 4 Hz

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following brain regions when stimulated via magnetic pulse reduces depression?

Select one:

- Right temporal
- Left prefrontal
- Right occipital
- Left temporal
- Right parietal

Explanation:

When using repetitive transcranial stimulation (rTMS) for the treatment of depression, placement of the coil is on the left dorsolateral prefrontal cortex (DLPFC). Research suggests that rTMS results in changes in regional brain activity and metabolism, and that applying rTMS at the left DLPFC can enhance this region’s connectivity with other regions that are crucial for regulating emotional processing.

Hardy S et al. Transcranial magnetic stimulation in clinical practice. BJPsych Advances. 2016

Nov;22(6):373-9.

The correct answer is: Left prefrontal

Question **17**

Not answered

Marked out of 1.00

Flag question

Which stages of sleep are considered as slow wave sleep?

Select one:

- Stage 1 and 2 n-REM sleep
- Stage 3 and 4 n-REM sleep
- Stage 0 and 1 n-REM sleep
- REM sleep
- Stage 2 and 3 n-REM sleep

REM sleep: The eyes undergo rapid movements, and there is a high level of brain activity. Non-REM sleep: There is reduced neuronal activity. Stage 3 and 4 of NREM are grouped as slow wave sleep because the slow delta waves predominate in these two stages.

The correct answer is: Stage 3 and 4 n-REM sleep

Question **18**

Not answered

Marked out of 1.00

Flag question

Which type of EEG trace is expected when eyes are closed during relaxation?

Select one:

- Gamma
- Beta
- Theta
- Alpha
- Delta

Explanation:

Alpha waves (frequency range 8-13 Hz) are highly rhythmic and constitute the dominant brain wave frequency of the normal eyes-closed awake EEG.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 998.

The correct answer is: Alpha

Question **19**

Not answered

Marked out of 1.00

Flag

1-2 Hz shaped wave pattern is seen in

Select one:

- Hepatic encephalopathy
- Huntington disease

question

- Alzheimer disease
- CJD
- Multi-infarct dementia

The classic EEG finding in CJD is generalized 1-2 Hz bi- or triphasic sharp- and slow-wave complexes. These may not appear until very late in the clinical course, if at all, and may require repeated testing. The earliest EEG finding is more often rhythmic generalized slowing.

The correct answer is: CJD

Question **20**

Not answered

Marked out of 1.00

Flag question

At resting membrane potential which ion is greatest in concentration inside the cell?

Select one:

- Chloride
- Calcium
- Sodium
- Magnesium
- Potassium

Explanation:

At resting membrane potential, there are more potassium ions inside and more sodium ions outside the plasma membrane. A sodium-potassium pump continually pumps sodium ions out of the cell and potassium ions into the cell.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. p. 87.

The correct answer is: Potassium

Question **21**

Not answered

Marked out of 1.00

Flag question

Orexin (hypocretin) neurons are dysfunctional in which of the following disorders?

Select one:

- Obesity
- Catatonia
- Chorea
- Anorexia
- Narcolepsy

Explanation:

Narcolepsy is characterised by excessive daytime sleepiness, the intrusion of sleep during periods of wakefulness, and abnormal REM sleep, including sleep-onset REM periods. Cataplexy, or loss of muscle tone triggered by emotions, may also be present. A clear neuropathological substrate has been

identified for narcolepsy with cataplexy: profound loss of orexin neurons in the lateral hypothalamus. Orexin neurons are involved in stabilising wakefulness through stimulating release of wake-promoting neurotransmitters (serotonin, norepinephrine, dopamine, acetylcholine, and histamine). Orexin also stabilises motor movements, allowing normal movement in the day when orexin levels are high and facilitating inhibition of motor movements at night, especially during REM sleep, when orexin level are low. When orexin levels are low in the daytime due to loss of orexin neurons, this destabilises motor movements during the daytime, allowing intrusions of motor inhibition and loss of muscle tone, known as cataplexy, during periods of wakefulness.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 435.

The correct answer is: Narcolepsy

Question **22**

Not answered

Marked out of 1.00

Flag question

A 3 month-old newborn undergoes awake EEG investigation to detect seizure activity, but no abnormalities are found. Which of the following waveforms will be prominent in the recording?

Select one:

- Delta and theta
- Spindle waves
- Alpha and beta
- Mu and lambda
- K complexes

Explanation:

With maturation, EEG activity gradually goes from a preponderance of irregular medium- to high-voltage delta activity (3.5Hz or less) in the tracing of the infant, to greater frequency and more rhythmic pattern. Rhythmic activity in the upper theta-lower alpha range (7 to 8 Hz) can be seen in posterior areas by early childhood, and, by mid-adolescence, the EEG essentially has the appearance of an adult tracing.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 999.

The correct answer is: Delta and theta

Question **23**

Not answered

Marked out of 1.00

Flag question

Which of the following associated with neuroendocrine changes during sleep is correct?

Select one:

- Testosterone decreases when sleep starts
- GH decreases when sleeping
- Cortisol shoots up in slow wave sleep
- During REM sleep melatonin decreases
- Prolactin decreases in early morning sleep

Melatonin regulates circadian rhythms. It has both synchronizing and phase-shifting properties in the regulation of biological rhythms. It reduces during REM sleep.

The correct answer is: During REM sleep melatonin decreases

Question **24**

Not answered

Marked out of 1.00

Flag question

Which of the following is a posterior pituitary hormone?

Select one:

- Melatonin
- Gonadotrophin
- Prolactin
- Testosterone
- Oxytocin

Explanation:

The vasopressor effects of posterior pituitary extracts were first described in 1895. Oxytocin (OT) and vasopressin (AVP) mRNAs are among the most abundant messages in the hypothalamus, being heavily concentrated in the magnocellular neurons of the paraventricular nucleus and the supraoptic nucleus of the hypothalamus, which send axonal projections to the neurohypophysis. These neurons produce OT and AVP (also known as antidiuretic hormone), which is released into the bloodstream, where these peptides act like hormones on peripheral targets.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 969.

The correct answer is: Oxytocin

Question **25**

Not answered

Marked out of 1.00

Flag question

Which of the following disorders is characterised by a predominant presence of triphasic waves in EEG?

Select one:

- Head injury
- Depressive pseudo-dementia
- Herpes simplex encephalitis
- Dementia of Alzheimer type
- Hepatic encephalopathy

Explanation:

Triphasic waves typically consist of generalised synchronous waves occurring in brief runs. They are 1.5-3.0 per second high-voltage slow waves, with each slow wave initiated by a blunt or rounded spike-like transient. Approximately one half of patients with triphasic waves have hepatic encephalopathy. The remainder have other toxic-metabolic encephalopathies.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 999-1000.

The correct answer is: Hepatic encephalopathy

Question **26**

Not answered

Marked out of 1.00

Flag question

Mrs. X is a patient with bipolar affective disorder with secondary amenorrhea. The investigation to be done is

Select one:

- Serum testosterone levels
- FSH and LH levels
- Thyroid levels
- Pregnancy test
- Serum prolactin levels

The most common cause of secondary amenorrhoea is pregnancy. Also, anticonvulsants used in bipolar disorder can reduce the efficacy of oral contraceptives.

The correct answer is: Pregnancy test

Question **27**

Not answered

Marked out of 1.00

Flag question

K complexes during sleep EEG are seen during

Select one:

- Stage 3 sleep
- Stage 4 sleep
- Stage 2 sleep
- Stage 1 sleep
- REM sleep

Explanation:

The large multicomponent K-complex and the rhythmic spindle are the hallmarks of NREM stage 2 sleep.

Ref: Ioannides AA, Liu L, Kostopoulos GK. The Emergence of Spindles and K-complexes and the role of the dorsal caudal part of the anterior cingulate as the generator of K-complexes. Frontiers in neuroscience. 2019 Aug 7;13:814.

The correct answer is: Stage 2 sleep

Question **28**

Not answered

Marked out of 1.00

Flag question

The brain region that acts as monitor of circadian rhythm is

Select one:

- Hippocampus
- Suprachiasmatic nucleus
- Locus ceruleus
- Nucleus accumbens
- Dorsal medial thalamus

Explanation:

The key neurobiological regulator of the circadian sleep system is the suprachiasmatic nucleus of the hypothalamus.

Ref: Smith PC, Mong JA. Neuroendocrine control of sleep. Neuroendocrine Regulation of Behavior. 2019:353-78.

The correct answer is: Suprachiasmatic nucleus

Question **29**

Not answered

Marked out of 1.00

Flag question

The levels of ACTH and CRF are increased and play an important role in the neuroendocrinology of

Select one:

- Autism
- Dementia
- Schizophrenia
- Depression
- Bipolar disorder

Explanation:

The normal stress response involves activation of the hypothalamus and a resultant increase in corticotropin-releasing factor (CRF), which in turn stimulates the release of adrenocorticotropic hormone (ACTH) from the pituitary. ACTH causes glucocorticoid release from the adrenal gland. In situation of chronic stress, hyperactivity of the hypothalamic-pituitary-adrenal (HPA) axis leads to excessive glucocorticoid release. Elevated HPA activity is one of the clearest links between depression and the biology of chronic stress. Hypercortisolaemia in depression suggests one or more of the following central disturbances: decreased inhibitory serotonin tone; increased drive from norepinephrine, acetylcholine (Ach), or corticotropin-releasing hormone (CHR); or decreased feedback inhibition from the hippocampus.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 270-271.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 395.

The correct answer is: Depression

Question **30**

Not answered

Marked out of 1.00

Flag question

Regarding the neuronal resting membrane potential, which of the following is true?

Select one:

- Hyperpolarise with moving out of chloride.
- Depolarise with rapid moving out of sodium.
- Repolarise with moving out of potassium.
- Repolarise with rapid cellular entry of sodium.
- Depolarise with rapid cellular entry of potassium.

Explanation:

An action potential, also referred to as a spike, is an all-or-none, rapid, transient depolarisation of the neuron's membrane. An action potential occurs when localised potentials within the neuron sum to bring the membrane voltage to its threshold potential, around -55mV. This triggers the opening of voltage-gated sodium channels, and the rapid influx of sodium ions brings the membrane potential to a positive value. The membrane potential is restored to its normal resting value by the delayed opening of voltage-gated potassium channels and by the closing of the sodium channels. A refractory period follows an action potential, corresponding to the period when the voltage-gated sodium channels are inactivated.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. p. 88.

The correct answer is: Repolarise with moving out of potassium.

Question **31**

Not answered

Marked out of 1.00

Flag question

There is a nocturnal surge in the level of growth hormones during which of the following stages of sleep?

Select one:

- After waking up
- All stages of sleep
- Stage 1 and 2 n-REM sleep
- Slow wave sleep
- REM sleep

Explanation:

Growth hormone is elevated in the earlier portions of a sleep bout - particularly in slow-wave sleep - and decreases in later sleep phases.

Ref: Smith PC, Mong JA. Neuroendocrine control of sleep. Neuroendocrine Regulation of Behavior. 2019:353-78.

The correct answer is: Slow wave sleep

Question **32**

Not answered

Marked out of 1.00

Flag question

Which of the following statement concerning REM sleep is INCORRECT?

Select one:

- EEG shows activity similar to awake state
- Rapid eye movements are noted
- Cardiac activity is similar to awake state
- Muscle tone is similar to awake state
- Penile erection occurs

Explanation:

REM sleep is characterised by faster activity on an EEG – similar to that seen during periods of

wakefulness – as well as distinct eye movements, and peripheral muscle paralysis and loss of muscle tone called atonia. During REM sleep, autonomic nervous system activity increases and produces an increased pulse rate, elevated blood pressure, raised intracranial pressure, increased cerebral blood flow, greater muscle metabolism, and, in boys as well as men, erections.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 413.

Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 379.

The correct answer is: Muscle tone is similar to awake state

Question **33**

Not answered

Marked out of 1.00

Flag question

Which of the following stages of sleep predominates in a neonate?

Select one:

- Stage 2 sleep
- Stage 3 sleep
- Stage 4 sleep
- Stage 1 sleep
- REM sleep

Explanation:

Neonates sleep 16-20 hours a day, with about 50% of that time spent in REM sleep.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 383.

The correct answer is: REM sleep

Question **34**

Not answered

Marked out of 1.00

Flag question

Which of the following is an innovative treatment with some beneficial effect in dystonia?

Select one:

- Vagal nerve stimulation
- Transcranial magnetic stimulation (rTMS)
- Leucotomy
- Behavioural reversal therapy
- Nerve ablation

Explanation:

Repetitive transcranial magnetic stimulation (rTMS) or transcranial direct current stimulation (tDCS) may modulate abnormal sensorimotor networks and restore motor control in dystonia.

Ref: Lizarraga KJ et al. Update on current and emerging therapies for dystonia. Neurodegenerative Disease Management. 2019 Jun;9(3):135-47.

The correct answer is: Transcranial magnetic stimulation (rTMS)

Question **35**

Not answered

Marked out of 1.00

Flag question

Infants have a dominant rhythm of which of the following EEG wave frequencies?

Select one:

- 3 Hz
- 10 Hz
- 13 Hz
- 40 Hz
- 7 Hz

Explanation:

With maturation, EEG activity gradually goes from a preponderance of irregular medium- to high-voltage delta activity (3.5Hz or less) in the tracing of the infant, to greater frequency and more rhythmic pattern. Rhythmic activity in the upper theta-lower alpha range (7 to 8 Hz) can be seen in posterior areas by early childhood, and, by mid-adolescence, the EEG essentially has the appearance of an adult tracing.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 999.

The correct answer is: 3 Hz

Question **36**

Not answered

Marked out of 1.00

Flag question

Generalised spike-wave discharges with decreased waves on photic stimulation are seen in EEG of patients with;

Select one:

- Myoclonic epilepsy
- Absence seizures
- Generalised seizures
- CJD
- Angelmann syndrome

Explanation:

Certain maneuvers can provoke EEG abnormalities in patients suspected of having epilepsy. Hyperventilation and photic stimulation by looking at a stroboscopic light, the two most commonly used techniques, may trigger diagnostic epileptiform discharges and even seizures in patients with a primary or genetic generalised epilepsy.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman’s Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 208.

The correct answer is: Generalised seizures

Question **37**

Not answered

REM rebound on cessation is seen with the use of which of the following?

Marked out of 1.00

Flag question

Select one:

- Zolpidem
- Modafinil
- SSRIs
- Methylphenidate
- Inhalants

Stimulants reduce the total sleep time by decreasing both REM sleep and Slow Wave Sleep. On cessation of stimulants (except modafinil), REM rebound occurs.

The correct answer is: Methylphenidate

Question **38**

Not answered

Marked out of 1.00

Flag question

Which one of the following ions is freely permeable across cell membranes?

Select one:

- Chloride ions
- Organic ions
- Sodium ions
- Potassium ions
- All of the listed options

Potassium ions-Relatively permeable Sodium ions-relatively impermeable Chloride ions-freely permeable Organic ions-relatively impermeable

The correct answer is: Chloride ions

Question **39**

Not answered

Marked out of 1.00

Flag question

Which one of the following exists as CNS neurotransmitter as well as a hormone?

Select one:

- Glucagon
- Serotonin
- Vasopressin
- Dopamine
- Thyroxine

Explanation:

The vasopressor effects of posterior pituitary extracts were first described in 1895. Oxytocin (OT) and vasopressin (AVP) mRNAs are among the most abundant messages in the hypothalamus, being heavily concentrated in the magnocellular neurons of the paraventricular nucleus and the supraoptic nucleus of the hypothalamus, which send axonal projections to the neurohypophysis. These neurons produce OT and AVP (also known as antidiuretic hormone), which is released into the bloodstream,

where these peptides act like hormones on peripheral targets.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 969.

The correct answer is: Vasopressin

Question **40**

Not answered

Marked out of 1.00

Flag question

The duration of normal sleep latency in healthy adults is

Select one:

- 15-20 minutes
- 30-40 minutes
- 0-5 minutes
- 90-100 minutes
- 50-60 minutes

Explanation:

Sleep latency is measured as the time to sleep onset. It is normally about 15-20 minutes. Sleep latency >30min can be indicative of insomnia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 439.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 421.

The correct answer is: 15-20 minutes

Question **41**

Not answered

Marked out of 1.00

Flag question

Which one of the following classes of drugs increases beta activity on EEG?

Select one:

- Lithium
- Anticonvulsants
- Benzodiazepines
- Antipsychotics
- Antidepressants

Explanation:

Benzodiazepines generate a significant amount of diffuse beta activity. Both benzodiazepines and barbiturates typically produce excessive beta (fast) activity, which may be a telltale sign of surreptitious drug use.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 999.

Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 210.

The correct answer is: Benzodiazepines

Question **42**

Not answered

Marked out of 1.00

Flag question

Which one of the following is characteristic of REM sleep?

Select one:

- Atonia
- Reduced recall of dreaming if awoken
- Decreased heart rate
- Abolition of tendon reflexes
- Upward ocular deviation with few or no movements

Explanation:

Because most people awakened during a REM episode recall that they were dreaming, physicians equate REM sleep with dreaming. Dreams that occur during REM sleep possess emotional or intellectual complexity, at least on a superficial level, and rich visual imagery that people readily recall. REM sleep is characterised by faster activity on an EEG – similar to that seen during periods of wakefulness – as well as distinct eye movements, and peripheral muscle paralysis and loss of muscle tone called atonia. During REM sleep, autonomic nervous system activity increases and produces an increased pulse rate, elevated blood pressure, raised intracranial pressure, increased cerebral blood flow, greater muscle metabolism, and, in boys as well as men, erections.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 379.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 413.

The correct answer is: Atonia

Question **43**

Not answered

Marked out of 1.00

Flag question

A 42-year-old man is a journalist. He goes to sleep at 10PM every night and wakes up at 530AM every morning. If he volunteers for a sleep study using a polysomnogram, the time from sleep onset to the first REM episode will be

Select one:

- 30 minutes
- 60 minutes
- 160 minutes
- 10 minutes
- 90 minutes

Explanation:

A typical night's sleep has four or five cycles of sequential stages, each lasting 90-110 min. The end of the first sleep cycle is marked by a brief period of arousal before the onset of REM sleep (at approximately 90 min). As the night progresses, the amount of time spent in delta sleep decreases, with consequent increase in REM sleep. Hence, the first REM period may last 5-10 min, while the last, just before waking, may last up to 40 min.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p.

437.

The correct answer is: 90 minutes

Question **44**

Not answered

Marked out of 1.00

Flag question

Peak cortisol levels in normal individuals are noted around

Select one:

- 12 AM Mid night
- 9 AM after breakfast
- 6 AM in the morning
- 12 PM Mid day
- 6 PM in the evening

A diurnal variation in ACTH and cortisol levels occurs in humans, with peak cortisol levels occurring around 6:00-7:00 AM.

The correct answer is: 6 AM in the morning

Question **45**

Not answered

Marked out of 1.00

Flag question

A 34-year-old patient who is taking sulpiride undergoes EEG investigation for detecting drug-related EEG changes. What would you expect to see?

Select one:

- decrease in delta activity
- new onset spike and waves
- decrease in alpha activity
- slowing of beta activity
- decrease in theta activity

Slowing of beta activity with an increase in alpha, theta and delta activity is seen with antipsychotics.

The correct answer is: slowing of beta activity

Question **46**

Not answered

Marked out of 1.00

Flag question

A 47-year-old woman presents with low mood and lethargy. She is noted to have a history of untreated hypothyroidism. Which of the following is inconsistent with a diagnosis of hypothyroidism?

Select one:

- Increased TSH levels in blood
- Cold intolerance
- Decreased libido
- Weight gain
- Tachycardia

Explanation:

The most common symptoms of hypothyroidism in adults are fatigue, lethargy, cold intolerance, weight gain, constipation, change in voice, and dry skin. Signs may also include sexual dysfunction and bradycardia. Primary hypothyroidism is defined by TSH concentrations above the reference range.

Ref: Chaker L et al. Hypothyroidism. Lancet. 2017 Sep 23;390(10101):1550-1562.

The correct answer is: Tachycardia

Question **47**

Not answered

Marked out of 1.00

Flag question

Regarding major depression, which of the following statements on neuroendocrine abnormalities is correct?

Select one:

- Large dose of cortisol exacerbates depressive symptoms
- 50% of patients show raised cortisol levels
- Low levels of corticotrophin releasing factor is found in CSF of patients
- Dehydroepinandrosterone levels are low
- Atrophy of adrenal cortex is seen in 20% of patients

Many depressed patients have a high level of cortisol in plasma and urine. It is now known that 50% of depressed patients exhibit hypercortisolemia.

The correct answer is: 50% of patients show raised cortisol levels

Question **48**

Not answered

Marked out of 1.00

Flag question

Mr. Y presents to the emergency department with a history of apathy and weight loss. On further investigations, he has hyponatraemia and hyperkalemia. What is his possible diagnosis?

Select one:

- Acute porphyria
- Cushing syndrome
- Cohn syndrome
- Diabetes insipidus
- Addison disease

Explanation:

Addison's disease (primary adrenal insufficiency) is characterised by chronic glucocorticoid and/or mineralocorticoid deficiency due to failure of the adrenal cortex. It should be suspected in all acutely or chronically ill patients with general fatigue or severe weakness and unexplained dehydration, hypotension, weight loss, fever, abdominal pain and hyperpigmentation. Key lab characteristics include hyponatremia, hyperkalemia, and (especially in children) hypoglycemia. Psychiatric symptoms include apathy, anhedonia, fatigue, and depressed mood.

Ref: Barthel A et al. An update on Addison's disease. Experimental and Clinical Endocrinology & Diabetes. 2019 Feb;127(02/03):165-75.

The correct answer is: Addison disease

Question **49**

Not answered

Marked out of 1.00

Flag question

Peak cortisol level in normal physiological states is seen at

Select one:

- 10 AM
- 4 AM
- 8 AM
- 6 AM
- 2 PM

A diurnal variation in ACTH and cortisol levels occurs in humans, with peak cortisol levels occurring around 6:00-7:00 AM.

The correct answer is: 6 AM

Question **50**

Not answered

Marked out of 1.00

Flag question

Which one of the following is not a characteristic feature of Kleine-Levin syndrome?

Select one:

- Episodic nature
- Hypersomnia
- Male predominance
- Hypertension
- Hyperphagia

Explanation:

Kleine-Levin syndrome is a rare syndrome of 'periodic somnolence and morbid hunger', occurring almost exclusively in male adolescents. It is characterised by periods lasting from days to weeks of attacks of hypersomnia, accompanied by excessive food intake.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 452.

The correct answer is: Hypertension

Question **51**

Not answered

Marked out of 1.00

Flag question

Which of the following receptor on stimulation leads to sleep onset?

Select one:

- Noradrenergic - receptors
- 5HT2 receptors
- Glucocorticoid receptors

- Histaminergic receptors
- Cholinergic receptors

The noradrenergic rapid eye movement (REM)-OFF neurons in locus coeruleus and cholinergic REM-ON neurons in tegmentum show a reciprocal firing pattern. The REM-ON neurons fire during REM sleep whereas REM-OFF neurons stop firing during REM sleep.
http://www.ncbi.nlm.nih.gov/pubmed/17704548

The correct answer is: Cholinergic receptors

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